

IFC5-X-ECS prototype

```
1  /accounts/<account_id>/projects/<project_id>/
2
3  /entities?is_version=A36BD7&has_component=IfcWallAttributes
4
5  /entities?is_version=D7A35AC&has_component=IfcPSetWallCommon => [ EntityId1, EntityId2, ...,
6      EntityIdN]
7
8  /entities/<entity_guid> => [ "ComponentAInstanceGUID", "ComponentBInstanceGUID" ]
9
10 /components/<component_guid> => {
11     "entity_guid": "entity_guid",
12     "instance_guid": "guidasdfsad",
13     "component_type": "uri",
14     "inherits_from": "other_component_guid",
15
16     "key1": "value1",
17     "key2": "value2",
18     ...,
19     "keyN": "valueN"
20 }
21 }
```

Type: IfcBuildingAttributesComponent

"instance_guid": "guidasdfsad",	Composition Type
"component_type": "uri",	Description
"entity_guid": "abcdefg",	Element Type
"properties": {	Global ID
	Long Name
	Name
	Object Type

Type: IfcAddressPsetComponent

"instance_guid": "guidasdfsad",	Electronic Mail Addresses
"component_type": "uri",	Facsimile Numbers
"entity_guid": "abcdefg",	Internal Location
"properties": {	Messaging Ids
	Pager Number
	Postal Box
	Postal Code
	Purpose
	Region
	Telephone Numbers

MyBuildingGUID

IfcBuildingAttributesComponent

SchemaUri: "https://..."
InstanceGUID: <guid>
EntityGUID: "MyBuildingGUID"
Description: "My Building Description"
Name: "123 Any Street"
Long Name: "Longer than Name"

IfcAddressPsetComponent

SchemaUri: "https://..."
InstanceGUID: <guid>
EntityGUID: "MyBuildingGUID"
Emails: ["a@b.com", "b@c.com"]
Postal Box: "PO 9887"
Purpose: "Mailing"

Principles

"Storage is not interesting"

Adopt git-like **component** versioning for:

- * provenance
- * branching
- * tagging

IFC5 will componentize the mesh format (not opinionated)

"git-like" versioning, no version IDs at the component level

Open world architecture: components on entities can exist anywhere

MVP

componentize 4.x "as is"

- Ifc[class]AttributeComponent
Adopt the class stack of attributes
- Ifc[Pset]PsetComponent
Adopt the full pset parameters
- Ifc[Relation]RelationComponent

Stand up reference implementation

Service: converter, query

Viewer: asks for components,
displays mesh format

JsonSchema of Ifc4.x converted to
component definitions

OpenAPI 3.1 definition of the API

Backlog / TBD

Ownership / Permissions

Versioning across stores in an open world

queries / selection in the API call
(probably GraphQL or SHACL)

"Layers", ie conflict resolution among
components / parameters

Cloud Events inspired transactions, ie
"what is the header on a transaction"

Two components of the same type:
override or augment?

IFC Processing Server

Upload IFC or JSON files for processing

Open Viewer

Server running • Version 0.1.0

Models in Data Store

Loaded from fileBased store

Refresh

Delete Selected

- | | |
|---|-----------------|
| ☐ HelloWall-01 | 36 components |
| ☐ HelloWall-02 | 49 components |
| ☐ HelloWall-03 | 59 components |
| ☐ RE1-Architecture | 757 components |
| ☐ RE1-Electrical | 615 components |
| ☐ RE1-Mechanical | 1108 components |

Select File



Click to select or drag and drop
Supported: .ifc, .json (Max 500MB)

Upload & Process

How it works:

- IFC files: Automatic conversion to JSON components

http://localhost:5000/api/components?models=RE1-Architecture,RE1-MEP,RE1-Electrical,RE1-Plu

Execute

Clear

✓ Loaded 3457 components (3 entities/models)

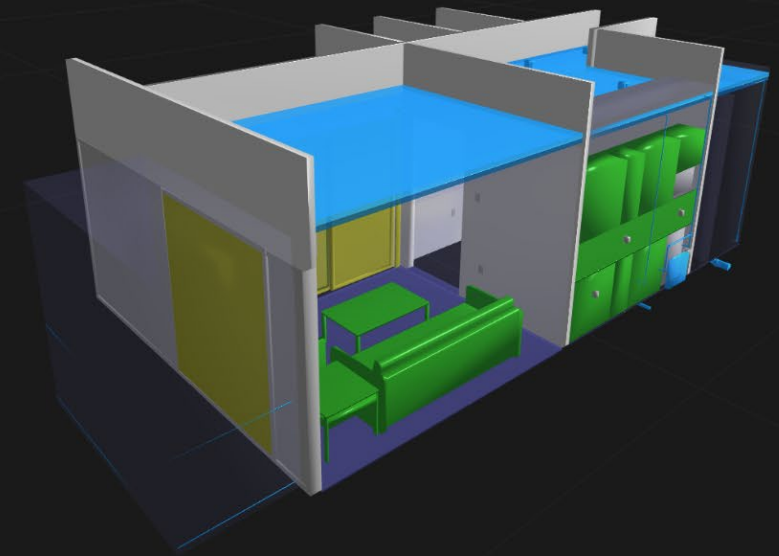
E/C

Info



- 0f06c449-9e40-e60b-4807-a4717... (1)
- 0f30f638-80b4-83a3-4a64-af2a39... (1)
- 10f9e5de-2705-9998-ba86-898d4... (2)
- 1179d690-1c44-a226-11e3-c77f5... (1)
- 12a73f69-32f1-4559-8fdb-39f6b3... (6)
 - Pset_QuantityTakeOff | f1066582-...
 - componentGuid: f1066582-756f-5eb1-...
 - componentType: IfcPropertySetCompo...
 - entityGuid: 12a73f69-32f1-4559-8fdb-3...
 - hasProperties: [1]
 - [0]: (3)
 - componentType: IfcPropertySin...
 - name: Reference
 - nominalValue: (2)
 - componentType: IfcIdentifi...
 - wrappedValue: -
 - name: Pset_QuantityTakeOff
 - d07c3f11-a87e-625d-e3fc-f6a26...
 - Pset_ReinforcementBarPitchOfW...
 - Pset_ConcreteElementGeneral | 1...
 - Basic Wall:-:418450 | 14d86059-c...
 - Pset_WallCommon | 24e06193-cf...
 - 12a73f69-32f1-4559-8fdb-39f6b3... (5)
 - Pset_QuantityTakeOff | 344f4c9f-...
 - Pset_SlabCommon | 9a5a54f6-02...
 - Pset_ReinforcementBarPitchOfSla...
 - 2ceb510d-dd2e-6560-b76d-7c22...
 - Floor:-:418660 | 58748c5d-f86a-...
 - 12a73f69-32f1-4559-8fdb-39f6b3... (4)
 - Pset_CoveringCommon | ff15754...
 - Compound Ceiling:-:418801 | 45...
 - Pset_QuantityTakeOff | 0c1b592e...

3D View



Grid Axes

Component data strategy

Class

Name

Code

Identifier (URI)

Definition

Description

Synonyms

Related IFC entities

Parent class

Child classes

Object

IfcObject

Door

IfcDoor

https://identifier.buildingsmart.org/uri/buildingsmart/ifc/4.3/class/ifcDoor

The door is a built element that is predominately used to provide controlled access for people, goods, animals and vehicles. It includes constructions with hinged, pivoted, sliding, and additionally revolving and folding operations.

Built Element

Boom Barrier, Door, Gate, Trapdoor, Turnstile

Properties (4)

Name	Data type
^ Attributes	
Description	String
Global ID	String
Name	String
Object Type	String

Properties (238)

Name	Data type	Units	Definition	Dictionary	Identifier (URI)
^ Attributes					
Description	String		Optional description, provided for exchanging informative comments.	IFC	
Element Type	String		The type denotes a particular type that indicates the object further. The use has to be established ...	IFC	
Global ID	String		Assignment of a globally unique identifier within the entire software world.	IFC	
Name	String		Optional name for use by the participating software systems or users. For some subtypes of IfcRoot...	IFC	
Object Type	String		The type denotes a particular type that indicates the object further. The use has to be established ...	IFC	
Operation Type	String		Type of operations. Also used to assign standard symbolic presentations according to national bu...	IFC	
Overall Height	Real		Overall measure of the height, it reflects the Z Dimension of a bounding box, enclosing the body of ...	IFC	
Overall Width	Real		Overall measure of the width, it reflects the X Dimension of a bounding box, enclosing the body of t...	IFC	

Class Definition with Instance parameters

Starting with:
Object definitions,
Property Sets,
Relations,
ShapeRepresentations,

...

^ Pset_DoorCommon					
Acoustic Rating	String	Acoustic rating for this object.; It is provided according to the national building code. It indicat...	IFC		
Durability Rating	String	Durability against mechanical stress. It is given according to the national code or regulation.	IFC		
Fire Exit	Boolean	Indication whether this object is designed to serve as an exit in the case of fire (TRUE) or not (FA...	IFC		
Fire Rating	String	Fire rating for this object. It is given according to the national fire safety classification.	IFC		
Glazing Area Fraction	Real	Fraction of the glazing area relative to the total area of the filling element.; It shall be used, i...	IFC		
Handicap Accessible	Boolean	Indication that this object is designed to be accessible by the handicapped. Set to (TRUE) if this o...	IFC		
Has Drive	Boolean	Indication whether this object has an automatic drive to operate it (TRUE) or no drive (FALSE)	IFC		
Hygrothermal Rating	String	Resistance against hygrothermal impact from different temperatures and humidities inside and outside...	IFC		
Infiltration	Real	Infiltration flowrate of outside air for the filler object based on the area of the filler objec...	IFC		
Is External	Boolean	Indication whether the element is designed for use in the exterior (TRUE) or not (FALSE). If (TRUE) ...	IFC		
Mechanical Load Rating	String	Mechanical load rating for this object.; It is provided according to the national building code.	IFC		
Security Rating	String	Index based rating system indicating security level.; It is giving according to the national buildin...	IFC		
Self Closing	Boolean	Indication whether this object is designed to close automatically after use (TRUE) or not (FALSE).	IFC		
Smoke Stop	Boolean	Indication whether the object is designed to provide a smoke stop (TRUE) or not (FALSE).	IFC		
Status	String	Status of the element, predominately used in renovation or retrofitting projects. The status can...	IFC		
Thermal Transmittance	Real	Thermal transmittance coefficient (U-Value) of an element, within the direction of the thermal f...	IFC		
Water Tightness Rating	String	Water tightness rating for this object.; It is provided according to the national building code.	IFC		

Property Sets

Data migration strategy

ClassType Component IfcDoor

Name	Data type	Units	Definition	Dictionary	Identifier (URI)
^ Attributes					
EntityGuid					
EntityType					
ComponentGuid					
ComponentType					
Description	String		Optional description, provided for exchanging informative comments.	IFC	📄 ▼
Element Type	String		The type denotes a particular type that indicates the object further. The use has to be established ...	IFC	📄 ▼
Global ID	String		Assignment of a globally unique identifier within the entire software world.	IFC	📄 ▼
Name	String		Optional name for use by the participating software systems or users. For some subtypes of IfcRoot...	IFC	📄 ▼
Object Type	String		The type denotes a particular type that indicates the object further. The use has to be established ...	IFC	📄 ▼
Operation Type	String		Type of operations. Also used to assign standard symbolic presentations according to national bu...	IFC	📄 ▼
Overall Height	Real		Overall measure of the height, it reflects the Z Dimension of a bounding box, enclosing the body of ...	IFC	📄 ▼
Overall Width	Real		Overall measure of the width, it reflects the X Dimension of a bounding box, enclosing the body of t...	IFC	📄 ▼

Propertyset Component IfcDoor

Name	Data type	Units	Definition	Dictionary	Identifier (URI)
^ Attributes					
EntityGuid					
ComponentGuid					
ComponentType					
^ Pset_DoorCommon					
Acoustic Rating	String		Acoustic rating for this object.; It is provided according to the national building code. It indicat...	IFC	📄 ▼
Durability Rating	String		Durability against mechanical stress. It is given according to the national code or regulation.	IFC	📄 ▼
Fire Exit	Boolean		Indication whether this object is designed to serve as an exit in the case of fire (TRUE) or not (FA...	IFC	📄 ▼
Fire Rating	String		Fire rating for this object. It is given according to the national fire safety classification.	IFC	📄 ▼
Glazing Area Fraction	Real		Fraction of the glazing area relative to the total area of the filling element.; It shall be used, i...	IFC	📄 ▼
Handicap Accessible	Boolean		Indication that this object is designed to be accessible by the handicapped. Set to (TRUE) if this o...	IFC	📄 ▼
Has Drive	Boolean		Indication whether this object has an automatic drive to operate it (TRUE) or no drive (FALSE)	IFC	📄 ▼

0d68ad35-29c7-484d-907c-65a3fbc0eadb
(IfcDoor)

Data migration strategy

0d68ad35-29c7-484d-907c-65a3fbc0eadb
(IfcDoor)

ClassType Component IfcDoor

Name	Data type	Units	Definition	Dictionary	Identifier (URI)
^ Attributes					
EntityGuid			<- V4 IFC ID		
EntityType			<- The V4 class name		
ComponentGuid			<- New Guid		
ComponentType			<- EntityType+Component		
Description	String		Optional description, provided for exchanging informative comments.	IFC	📄 ▼
Element Type	String		The type denotes a particular type that indicates the object further. The use has to be established ...	IFC	📄 ▼
Global ID	String		Assignment of a globally unique identifier within the entire software world.	IFC	📄 ▼
Name	String		Optional name for use by the participating software systems or users. For some subtypes of IfcRoot...	IFC	📄 ▼
Object Type	String		The type denotes a particular type that indicates the object further. The use has to be established ...	IFC	📄 ▼
Operation Type	String		Type of operations. Also used to assign standard symbolic presentations according to national bu...	IFC	📄 ▼
Overall Height	Real		Overall measure of the height, it reflects the Z Dimension of a bounding box, enclosing the body of ...	IFC	📄 ▼
Overall Width	Real		Overall measure of the width, it reflects the X Dimension of a bounding box, enclosing the body of t...	IFC	📄 ▼

Propertyset Component IfcDoor

Name	Data type	Units	Definition	Dictionary	Identifier (URI)
^ Attributes					
EntityGuid			<- Guid of Entity		
ComponentGuid			<- V4 IFC ID		
ComponentType			<- IFCPropertySet		
^ Pset_DoorCommon					
Acoustic Rating	String		Acoustic rating for this object.; It is provided according to the national building code. It indicat...	IFC	📄 ▼
Durability Rating	String		Durability against mechanical stress. It is given according to the national code or regulation.	IFC	📄 ▼
Fire Exit	Boolean		Indication whether this object is designed to serve as an exit in the case of fire (TRUE) or not (FA...	IFC	📄 ▼
Fire Rating	String		Fire rating for this object. It is given according to the national fire safety classification.	IFC	📄 ▼
Glazing Area Fraction	Real		Fraction of the glazing area relative to the total area of the filling element.; It shall be used, i...	IFC	📄 ▼
Handicap Accessible	Boolean		Indication that this object is designed to be accessible by the handicapped. Set to (TRUE) if this o...	IFC	📄 ▼
Has Drive	Boolean		Indication whether this object has an automatic drive to operate it (TRUE) or no drive (FALSE)	IFC	📄 ▼

ClassTypeComponent

```
{
  "componentGuid": "5c890f20-ee21-063a-7d35-3cffdf73b494",
  "componentType": "IfcDoorComponent",
  "entityGuid": "f1c8fc3e-82ee-4c75-ae69-2b1be0926228",
  "entityType": "IfcDoor",
  "name": "Door-Passage-Single-Flush:36\" x 84\":1240792",
  "objectType": "Door-Passage-Single-Flush:36\" x 84\"",
  "overallHeight": 7.333333333333333,
  "overallWidth": 3.333333333333333,
  "tag": "1240792"
}
```

PropertySetComponent

```
{
  "componentGuid": "6d533265-65f4-fc00-e847-7e73cc3ab3f7",
  "componentType": "IfcPropertySetComponent",
  "entityGuid": "f1c8fc3e-82ee-4c75-ae69-2b1be0926228",
  "hasProperties": [
    {
      "componentType": "IfcPropertySingleValueComponent",
      "name": "Reference",
      "nominalValue": {
        "componentType": "IfcIdentifierComponent",
        "wrappedValue": "36\" x 84\""
      }
    }
  ],
  "name": "Pset_QuantityTakeOff"
}
```

ClassTypeComponent

```
{
  "componentGuid": "5c890f20-ee21-063a-7d35-3cffdf73b494",
  "componentType": "IfcDoorComponent",
  "entityGuid": "f1c8fc3e-82ee-4c75-ae69-2b1be0926228",
  "entityType": "IfcDoor",
  "name": "Door-Passage-Single-Flush:36\" x 84\":1240792",
  "objectType": "Door-Passage-Single-Flush:36\" x 84\"",
  "overallHeight": 7.333333333333333,
  "overallWidth": 3.333333333333333,
  "tag": "1240792"
}
```

ShapeRepresentationComponent

```
{
  "componentGuid": "cbb1b9d8-b5b8-d604-1715-c5157acc3229",
  "componentType": "IfcShapeRepresentationComponent",
  "entityGuid": "f1c8fc3e-82ee-4c75-ae69-2b1be0926228",
  "representationIdentifier": "Body",
  "representationFormat": "OBJ",
  "items": [
    "v -2.01486169281188 0.023758 1.0
    f 13 2 3\nf 13 3 4\nf 13 4 5\nf 13 5 6\nf 13 7 8\n"
  ]
}
```

Relations

```
{
  "entityGuid": "f1c8fc3e-82ee-4c75-ae69-2b1be0926228",
  "componentGuid": "933c4a06-93b8-11d3-80f8-00c14f8efc2c",
  "componentType": "IfcRelContainedInSpatialStructureComponent",
  "relatedElements": [
    "0d68ad35-29c7-484d-907c-65a3fbc0eadb"
  ],
  "relatingStructure": "933c4a06-93b8-11d3-80f8-00c04f8efc2c"
},
```

by reference
if GlobalId

```
{
  "entityGuid": "f1c8fc3e-82ee-4c75-ae69-2b1be0926228",
  "componentGuid": "5ee46b9f-a508-fc7b-3fde-e8df6c667753",
  "componentType": "IfcRelAssociatesMaterialComponent",
  "relatedObjects": [
    "0d68ad35-29c7-484d-907c-65a3fbc0eadb"
  ],
  "relatingMaterial": {
    "componentType": "IfcMaterialLayerSetUsageComponent",
    "directionSense": "NEGATIVE",
    "forLayerSet": {
      "componentType": "IfcMaterialLayerSetComponent",
      "layerSetName": "Basic Wall:Generic - 8\\\"",
      "materialLayers": [
        {
          "componentType": "IfcMaterialLayerComponent",
          "layerThickness": 0.6666666666666666,
          "material": {
            "componentType": "IfcMaterialComponent",
            "name": "Default Wall"
          }
        }
      ]
    }
  },
  "layerSetDirection": "AXIS2",
  "offsetFromReferenceLine": 0.3333333333333333
},
```

by value

Basic API

End points include:

<code>/models</code>	<code>#lists the models on the server</code>
<code>/entityTypes</code>	<code>#lists the types of entities in the model</code>
<code>/entityGuids</code>	<code>#lists the entityGuids</code>
<code>/componentGuids</code>	<code>#lists the componentGuids</code>
<code>/components</code>	<code>#returns component data</code>

Wherever appropriate the following **filters** can be applied:

```
?models=model1,model2,...
?entityTypes=IfcType1,IfcType2,...
?componentTypes=IfcType1Component,IfcType2Component,...
?entityGuids=guid1,guid2,...
?componentGuids=guid1,guid2,...
```

Component and entity type filters
observe IFC class hierarchy

Example queries:

```
http://localhost:5000/api/entityGuids?models=HelloWall-01,HelloWall-02
```

```
http://localhost:5000/api/componentGuids?entityTypes=IfcProduct
```

```
http://localhost:5000/api/components?componentTypes=IfcRelAggregatesComponent
```

```
http://localhost:5000/api/components?componentTypes=IfcRelAggregatesComponent
```

```
http://localhost:5000/api/components?componentGuids=156d4a9b-8c6b-d7dd-d6fc-b0dabd0b3d9c,8fcce169-98e4-9471-53dd-a583112814c7
```

Import simplifications

should maybe be in a .JSON config file

```
# include empty properties in output (as empty strings)
```

```
INCLUDE_EMPTY_PROPERTIES = False
```

```
#only these top level nodes are exported, all are imported
```

```
ALLOWED_TYPES = {  
    'IfcObjectDefinition',  
    'IfcPropertySet',  
    'IfcRelationship'  
}
```

```
# Define attributes to exclude
```

```
EXCLUDE_ATTRIBUTES = {  
    'ownerhistory',  
    'id',  
    'step_id',  
    'objectplacement',  
    'representation',  
    'representations',  
    'representationmaps',  
    'representationcontexts',  
    'unitsincontext',  
    'globalId'  
}
```

```
# Define attribute name substitutions
```

```
ATTRIBUTE_SUBSTITUTIONS = {  
    'Name': 'componentName',  
    'Description': 'componentDescription',  
    'HasPropertySets': 'propertySets',  
    'type': 'componentType'  
}
```

All attribute names converted to lowerCamelCase

File store (*Git'able, not sure how yet*)

IFC Processing Server

Upload IFC or JSON files for processing

[Open Viewer](#)

Server running • Version 0.1.0

Models in Data Store
Loaded from fileBased store

[Refresh](#) [Delete Selected](#)

☐ HelloWall-01

36 components

☐ HelloWall-02

49 components

☐ HelloWall-03

59 components

☐ RE1-Architecture

757 components

☐ RE1-Electrical

615 components

☐ RE1-Mechanical

1108 components

Select File

Click to select or drag and drop

Supported: Ifc, json (Max 500MB)

[Upload & Process](#)

Success

File processed successfully! ✓

Entities processed: 36

Components stored: 36

Store path:
C:_LOCAL\Git\IFCPrototype\server\dataStores\fileBased\data\HelloWall-01

How it works:

- IFC files: Automatic conversion to JSON components
- JSON files: Direct storage of components
- Components stored in dataStores/fileBased/data/
- Each component saved as entityGuid_guid.json

>HelloWall-01

server > dataStores > fileBased > data > HelloWall-01

Search HelloWall-01

New

Sort

View

Details

Name	Date modified	Type	Size
0d68ad35-29c7-484d-907c-65a3fbc0eadb_5d7be3a6-fe8b-0924-729b-3...	2/16/2026 6:39 PM	JSON Source File	
0d68ad35-29c7-484d-907c-65a3fbc0eadb_70c8c03a-adc2-63f9-182c-1...	2/16/2026 6:39 PM	JSON Source File	
0fef9ff3-3118-a8fe-7057-674186238e33_4f12d893-de9d-c7f4-fb08-44f3...	2/16/2026 6:39 PM	JSON Source File	
1feef250-04e2-de8f-37a7-5bf9bce301d3_d0383b75-4697-e391-67f6-00b...	2/16/2026 6:39 PM	JSON Source File	
4d85f1ec-595b-0ac2-499f-afc4d1346899_f11a4f2a-c1cf-3371-dc90-c57...	2/16/2026 6:39 PM	JSON Source File	
5ee46b9f-a508-fc7b-3fde-e8df6c667753_04d6a687-5166-5272-680c-4b...	2/16/2026 6:39 PM	JSON Source File	
6e91c622-bea2-2ccb-1c88-b29bfd516fd_24f8178b-1c86-76bf-1cfe-2e...	2/16/2026 6:39 PM	JSON Source File	
7b8f559f-8b31-42c2-ba44-e16f7a04146e_2175f3de-a957-ce26-6954-c9c...	2/16/2026 6:39 PM	JSON Source File	
7cacd49c-ac17-4591-ad0a-cbcabb412ad9_e680a5c5-2214-8009-0f18-c...	2/16/2026 6:39 PM	JSON Source File	
8fcce169-98e4-9471-53dd-a583112814c7_eb5dd21f-6f4f-065a-2b0c-67...	2/16/2026 6:39 PM	JSON Source File	
9f9e9729-baaf-50ca-3b3e-0dcabd388fc0_d9dfc0c9-f733-2492-00f5-bef...	2/16/2026 6:39 PM	JSON Source File	
38ad01c4-e491-2658-6ea0-a5b280c7191e_cc330b92-1fed-1f22-37d9-49...	2/16/2026 6:39 PM	JSON Source File	
156d4a9b-8c6b-d7dd-d6fc-b0dabd0b3d9c_1f47ee26-bde5-f07f-53a5-8...	2/16/2026 6:39 PM	JSON Source File	
710c4a31-8030-e3f8-36b4-099c01bad6a3_ee011be3-ad1c-ae20-7e02-64...	2/16/2026 6:39 PM	JSON Source File	
933c4a06-93b8-11d3-80f8-00c04f8efc2c_fb7d09b0-95ac-71a8-806f-146...	2/16/2026 6:39 PM	JSON Source File	
933c4a06-93b8-11d3-80f8-00c14f8efc2c_fea464e8-1e28-8717-de16-908...	2/16/2026 6:39 PM	JSON Source File	
951efe5d-db7c-52a2-8f27-a7af698e355b_4b881c84-4b80-12c4-013a-6c...	2/16/2026 6:39 PM	JSON Source File	

Box

iCloud Drive

OneDrive

Rensselaer Polytechnic

This PC

Local Disk (C:)

T7 (D:)

Google Drive (G:)

36 items

IFC-ECS vs. IFCX-alpha

“likely compatible”

IFC-ECS	IFCX-alpha
V4.X and simple EC inspired	USD inspired
Simplest questions first	Hardest questions first
API and database first	File first
so far only entityGuids, componentGuids, and components	Lots more: “classes”, “layers”, ...
Global placement	Localized placement

IFC-ECS vs. IFCX-alpha

“likely compatible”

```
<> viewer.html  DEPLOYMENT.md 9+  HelloWall-02.json  ...
C: > _LOCAL > GitHub > IFCXPrototype > data > HelloWall > {} HelloWall-02.json > ...

225 {
246 },
247   "name": "Pset_BuildingCommon"
248 },
249 {
250   "componentGuid": "99f2449e-e95b-991e-3500-be673cd26bd8",
251   "componentType": "IfcDoorComponent",
252   "entityGuid": "f1c8fc3e-82ee-4c75-ae69-2b1be0926297",
253   "entityType": "IfcDoor",
254   "name": "Door-Passage-Single-Flush:36\" x 84\":1240679",
255   "objectType": "Door-Passage-Single-Flush:36\" x 84\"",
256   "overallHeight": 7.333333333333333,
257   "overallWidth": 3.333333333333333,
258   "tag": "1240679"
259 },
260 {
261   "componentGuid": "9decc1fa-72e6-2a2e-9c7d-f12ba54cd5c1",
262   "componentType": "IfcWallStandardCaseComponent",
263   "entityGuid": "f1c8fc3e-82ee-4c75-ae69-2b1be09262cd",
264   "entityType": "IfcWallStandardCase",
265   "name": "Basic Wall:Generic - 8\":1240637",
266   "objectType": "Basic Wall:Generic - 8\"",
267   "tag": "1240637"
268 },
269 {
270   "componentGuid": "95f1f988-9176-a707-4ad3-ab37b22ccbdf",
271   "componentType": "IfcOpeningElementComponent",
272   "entityGuid": "dfbcf82b-f08c-b71f-8154-9b43694ff068",
273   "entityType": "IfcOpeningElement",
274   "name": "Door-Passage-Single-Flush:36\" x 84\":1240679:1"
```

```
{} hello-wall.ifcx.json  ...
C: > _LOCAL > GitHub > IFC5-development > Hello Wall > {} hello-wall.ifcx.json > {} 58 > {} attributes

840 {
843   "attributes": {
844     "UsdShade:MaterialBindingAPI": {
848     }
849   },
850 },
851 {
852   "def": "over",
853   "name": "N592504dc469a44d69ae8c801b591679b",
854   "attributes": {
855     "xformOp": {
856       "transform": [
857         [1,0,0,0],
858         [0,1,0,0],
859         [0,0,1,0],
860         [4.94660663604736,0,1,1]]
861     }
862   },
863 },
864 {
865   "def": "over",
866   "name": "N52836ca941db45b3ab06b0f6dafa28ed",
867   "attributes": {
868     "ifc5:spaceboundary": {
869       "relatedElement": {
870         "ref": "</N93791d5d5beb437bb8ec2f1f0ba4bf3b>"
871       }
872     }
873   }
```

Drshelden/IFCXPrototype: An ECS flavor of IFCX using simple IFC 4.X inspired components

github.com/Drshelden/IFCXPrototype/tree/main

Drshelden / IFCXPrototype

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About

Deployment Bot documentation 7e90491 · 12 hours ago 25 Commits

data	documentation	12 hours ago
docs	documentation	12 hours ago
server	documentation	12 hours ago
LICENSE	Initial commit	last week
README.md	small README.md fix	last week
requirements.txt	Post documentation	3 days ago

README MIT license

IFCXPrototype

An ECS flavor of IFCX using simple IFC 4.X inspired components

Live Demo

The full repo is <https://github.com/Drshelden/IFCXPrototype>.

A live version of the server is located on Heroku: <https://ifc-ecs-prototype-daa447b4d672.herokuapp.com/>

Queries can be run against this instance, for example:

About

An ECS flavor of IFCX using simple IFC 4.X inspired components

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Contributors 2

dennisrshelden Dennis Shelden

Drshelden

Languages

Python 55.3%

HTML 44.4%

Backlog

topics identified but not yet attempted

Entity and component names and name spaces. URI's, name registries, etc. Could be BSDD

Schema modularization. Json Schema provides useful mechanisms for partitioning and merging

Types and typicals, local placement. IFCXalpha treats this rigorously.

Conflict resolution. Layers...or not.

Git-like functionality has been proposed as the means for versioning, permissions, “work sets”, etc.

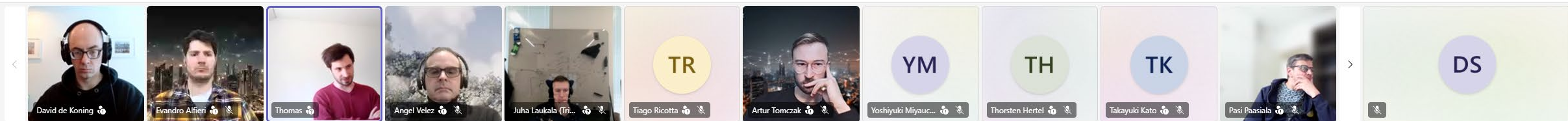
Transmission details – metadata wrappers for transmitted packages. [Cloud events](#) is a precedent.

Class hierarchy and parameter simplification. Can be done incrementally

External referencing of useful 3rd party ontologies, such as QUDT, etc.

Alternative encodings beyond Json and REST, JSON-LD, RDF, Graph-QL, SHACL, etc.

Indeterminate others...



Quick wins

ifc-ecs

json

ecs-like composition over inheritance

prioritize explicit geometry (obj literal)

ifc element taxonomy (componentType)

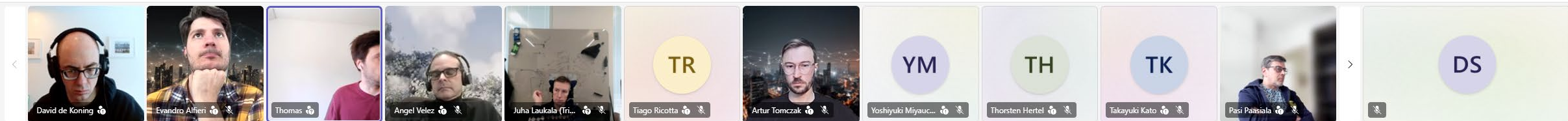
ifcx

json

ecs-like composition over inheritance

prioritize explicit geometry (usd-like triangle mesh array with semantic overlays)

ifc element taxonomy (bsi::ifc::class)



Discussion points

Serialization-level intelligence

ifc-ecs is more like a reserialization of IFC4

ifcx is more like a simplified JSON encoding of USD:

- composition across 'layers' for harmonizing decentralized content without API
- reducing schema complexity
 - type-objects, property sets -> inheritance
 - local placement, aggregation -> children
 - this is more or less a minimum if you want USD compatibility
 - this combination gives us reuse of aggregation trees (a limitation of IFC4 currently)